F 12 4th CAR

Lecture 26: Section 4.2
The Unit Circle and Trigonometric Functions

Unit circle

Derivation of the coordinates on the unit circle

Trigonemtric functions

Domain, range, and periods of trigonometric functions

Even-odd properties

Reminder:

Even functions: $f(-x) = f(x)$

odd functions: $f(-x) = -f(x)$

Lecture 26: Section 4.2
The Unit Circle and Trigonometric Functions

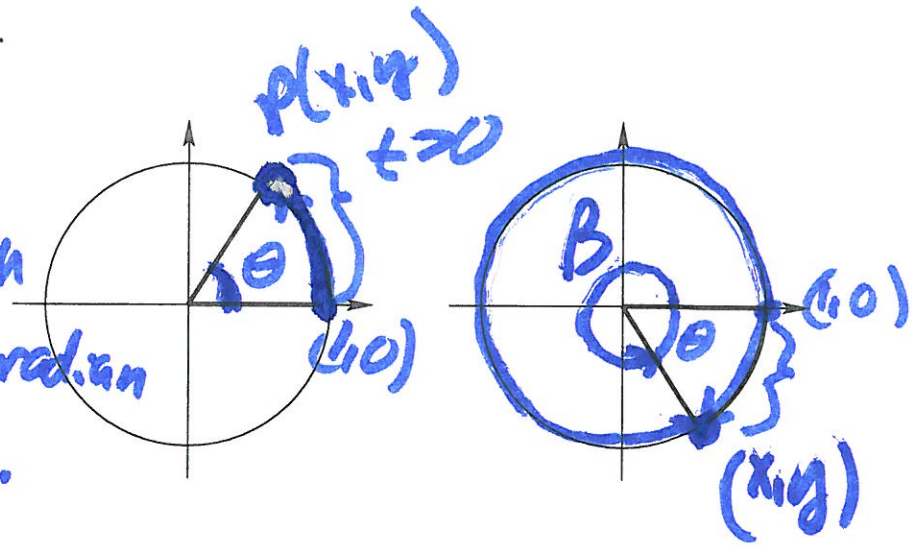
radius is 1

The **unit circle** is the circle of radius 1 centered at the origin in the xy-plane. Its equation is $x^2+y^2=1$.

Let t be a real number. The **terminal point** $P(x,y)$ on the unit circle is obtained by starting at the point (1,0) and moving in a counterclockwise direction with a distance t if $t>0$ or in a clockwise direction if $t<0$.

If radius=1

then arc length = the angle radian measurement.



$s = r\theta$

$s = 1(\theta)$

$s = \theta$

NOTE: All coterminal angles have the same terminal point.

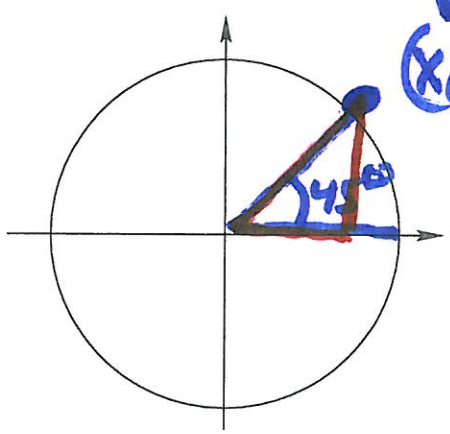
Derivation of coordinates on a unit circle.

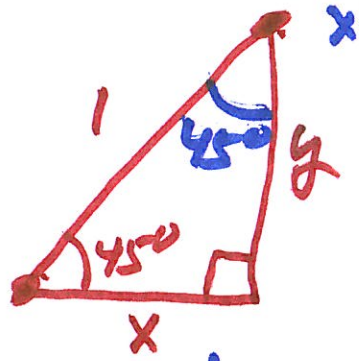
$\frac{\pi}{4} = \frac{180^\circ}{4} = 45^\circ$

ex. Find the terminal point on the unit circle corresponding to the real number t.

1) $t = \frac{\pi}{4}$ $= \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

(x, y)





$x^2 + y^2 = 1$

$x^2 + x^2 = 1$

$2x^2 = 1$

$x^2 = \frac{1}{2}$

$x = \pm\sqrt{\frac{1}{2}}$

isoceles Δ so $x = y$

$a^2 + b^2 = c^2$

$x = \pm\frac{\sqrt{2}}{2}$

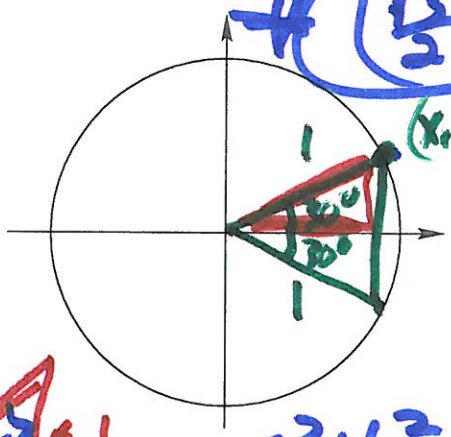
$x = \frac{\sqrt{2}}{2}$

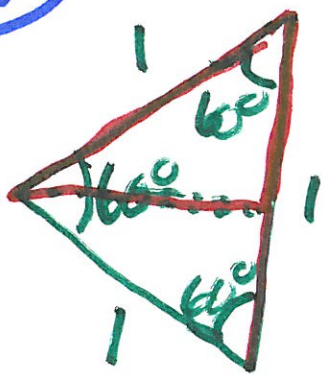
$y = \frac{\sqrt{2}}{2}$

2) $t = \frac{\pi}{6}$ $= 30^\circ$

$= \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

(x, y)



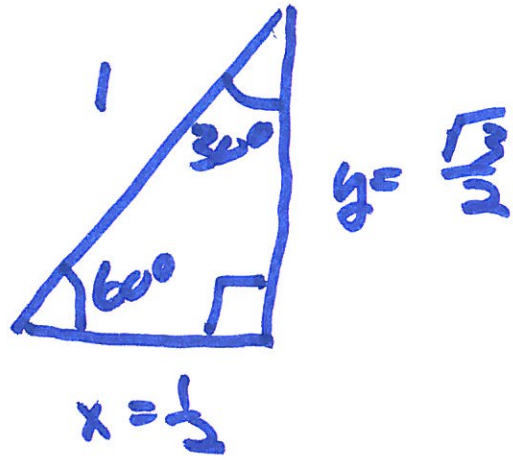


$a^2 + b^2 = c^2$

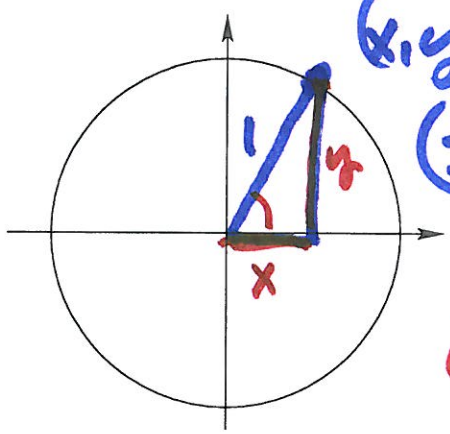
$x^2 + \left(\frac{1}{2}\right)^2 = 1^2$

$x^2 = \frac{3}{4}$

$x = \pm\sqrt{\frac{3}{4}} = \pm\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$



3) $t = \frac{\pi}{3} = 60°$



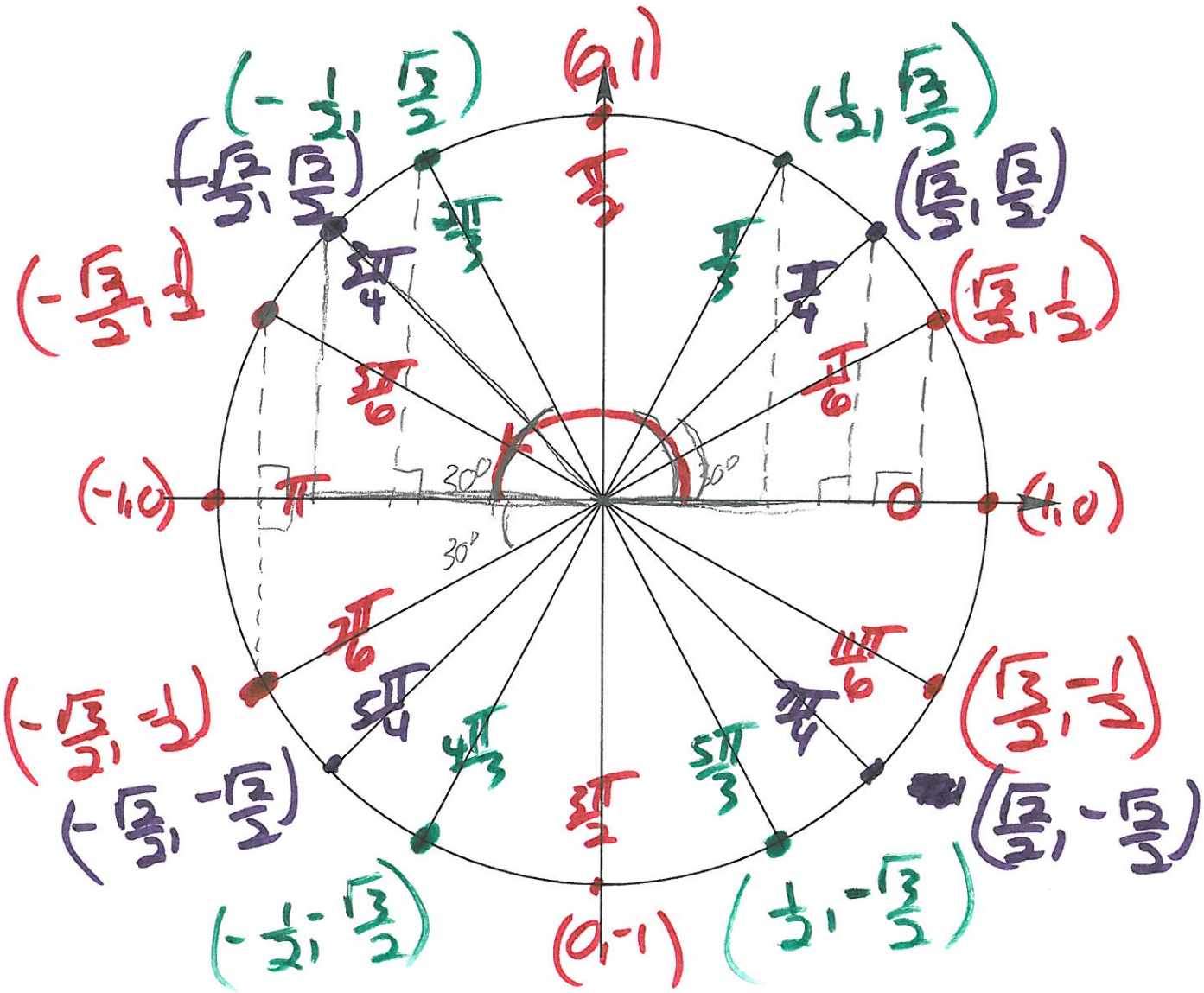
(x, y)

$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ *

60° x is smaller than y

$\frac{1}{2} < \frac{\sqrt{3}}{2}$

The Unit Circle



The Trigonometric Functions

Def. Let t be any real number and let (x, y) be the terminal point on the unit circle corresponding to t. We define

$$\sin t = y \qquad \csc t = \frac{1}{y}$$

$$\cos t = x \qquad \sec t = \frac{1}{x}$$

$$\tan t = \frac{y}{x} \qquad \cot t = \frac{x}{y}$$

ex. Find the six trigonometric functions of the given real number t.

$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

1) $t = \dfrac{\pi}{3}$

$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$ $\qquad \csc\frac{\pi}{3} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{2\sqrt{3}}{3}$

$\cos\frac{\pi}{3} = \frac{1}{2}$ $\qquad \sec\frac{\pi}{3} = \frac{1}{\frac{1}{2}} = 2$

$\tan\frac{\pi}{3} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$ $\qquad \cot\frac{\pi}{3} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{\sqrt{3}}{3}$

$(0,1)$

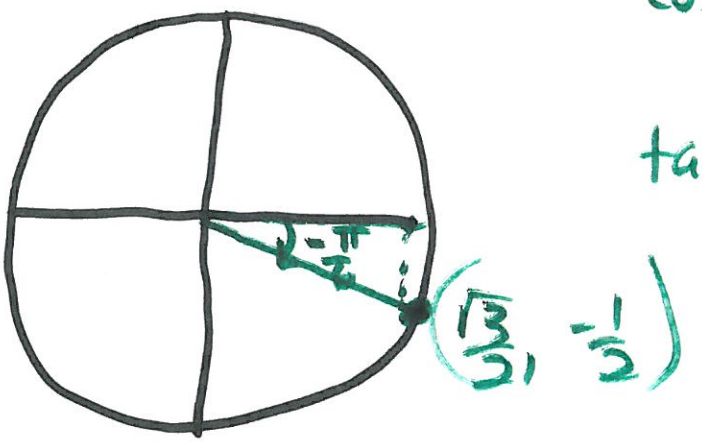
2) $t = \dfrac{\pi}{2}$

$\sin\frac{\pi}{2} = 1$ $\qquad \csc\frac{\pi}{2} = \frac{1}{1} = 1$

$\cos\frac{\pi}{2} = 0$ $\qquad \sec\frac{\pi}{2} = \frac{1}{0} =$ undefined

$\tan\frac{\pi}{2} = \frac{1}{0} =$ undefined $\qquad \cot\frac{\pi}{2} = \frac{0}{1} = 0$

3) $t = -\frac{\pi}{6}$



$\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$

$\cos\left(-\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$

$\tan\left(-\frac{\pi}{6}\right) = \frac{-\frac{1}{2}}{\frac{\sqrt{3}}{2}} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$

$\csc\left(-\frac{\pi}{6}\right) = -2$

$\sec\left(-\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

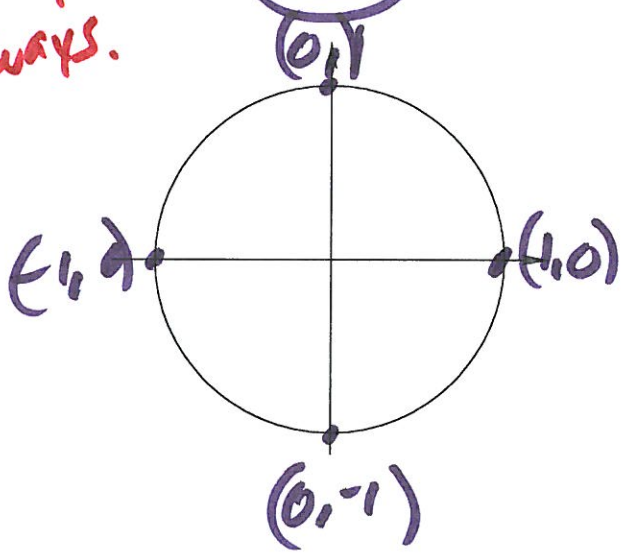
$\cot\left(-\frac{\pi}{6}\right) = -\sqrt{3}$

Domain and Range of Sine and Cosine

Domain: all real numbers $y = \sin x$

* Always.

Range: $[-1, 1]$



Periodic Functions

Def. A function f is **periodic** if there is a positive number c such that

$$f(t+c) = f(t)$$

for all t in the domain of f. The smallest number c satisfying the above equation is called the **period** of f.

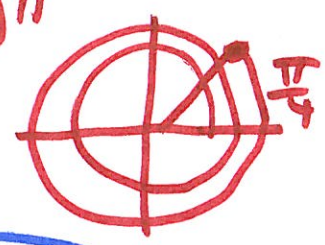
We have

$$\sin(t + 2\pi n) = \sin t \quad \text{and}$$

$$\cos(t + 2\pi n) = \cos t$$

for any integer n and real number t. The periods of sine and cosine functions are both 2π.

$\frac{17\pi}{4} = (4\frac{1}{4})\pi$

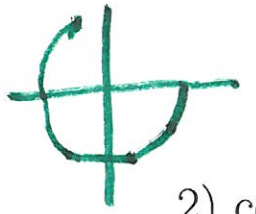


ex. Find → find where this is on the unit circle.

1) $\sin\left(\frac{17\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$\frac{17\pi}{4} - 2\pi = \frac{9\pi}{4}$

$\frac{9\pi}{4} - 2\pi = \frac{\pi}{4}$



2) $\cos\left(-\frac{4\pi}{3}\right)$

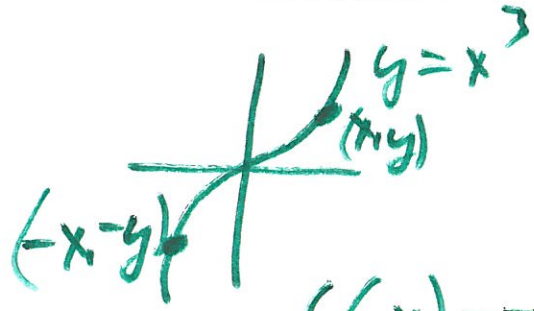
$-\frac{4\pi}{3} + 2\pi = \frac{2\pi}{3}$

$= -\frac{1}{2}$

even func. $y = x^2$

$f(-x) = f(x)$

odd functions



$f(-x) = -f(x)$

✳ Even-Odd Properties

1. The cosine and secant functions are even.

$$\cos(-t) = \cos t \qquad \sec(-t) = \sec t$$

2. The sine, cosecant, tangent, and cotangent functions are odd.

$$\sin(-t) = -\sin t \qquad \csc(-t) = -\csc t$$

$$\tan(-t) = -\tan t \qquad \cot(-t) = -\cot t$$

<u>ex.</u> Find

1) $\sin\left(-\frac{\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}$

2) $\cos\left(-\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

Practice.

Determine if the function is even, odd, or neither.

1) $\dfrac{\sin x}{x}$

2) $x \cos x$

3) $|x| + \tan x$

Answer. 1) even 2) odd 3) neither